

Client-Side Scripting II



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Bachelors of Science (Hons) (MQF Level 6)

Chapter 7 – Authentication

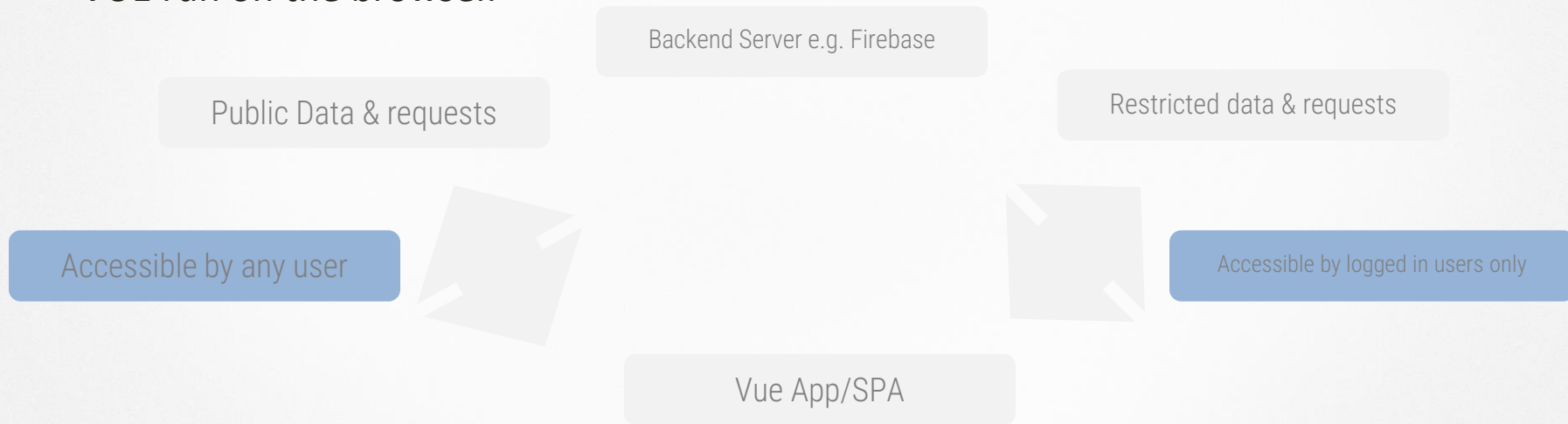
Contents

- ▶ How authentication works in Single Page Applications
- ▶ Locking backend data
- ▶ Firebase Authentication
- ▶ Firebase Auth REST API



How authentication works in SPAs

- ▶ Authentication in SPAs is a bit different then traditional authentication used in websites developed using Server-side languages since client-side languages such as VUE run on the browser.



How authentication works in SPAs

- ▶ A SPA is detached from the backend since it is not served from the same server and in such scenarios the server **does NOT care about the client** i.e. Our VUE app.
- ▶ Any server/API (in our case Firebase) **doesn't store any information** about the app or whether the current user of this app instance is logged in or not.
- ▶ The process always start by some **login data being sent to the server** e.g. username and password.
- ▶ In a traditional website the server would store information of that user through a session and then **send a session id to the frontend**/browser.



How authentication works in SPAs

- ▶ However in a SPA the server doesn't store any information **BUT sends a token to the frontend** in our case to the VUE app once login data is validated and verified.
- ▶ A token is just a **very long string which holds a secret key** generated by the server. Only the server knows this key and hence this token can't be fake.
- ▶ This token is then stored in VUEX or in permanent places like local storage.
- ▶ This token can then be **attached to the request** and since only the server can validate the token the server will **grant access and return the appropriate data**. If token is invalid or missing access is denied.

Locking backend data

- ▶ Login to Firebase with your google account.
- ▶ Create a new project. No need to add google Analytics
- ▶ Go to Realtime Database and create in test mode. Change rules to avoid time-limit but everyone will be able to access this data.

Authentication ▾

Realtime Database

Data

Rules

Backups

Usage

Edit rules

Monitor rules

```
1 {  
2   "rules": {  
3     ".read": true,  
4     ".write": true  
5   }  
6 }
```

Locking backend data

- ▶ The following can now be used as a backend to which we can send requests from inside our VUE app.

 <https://authentication-98e29-default-rtdb.europe-west1.firebaseio.com/>

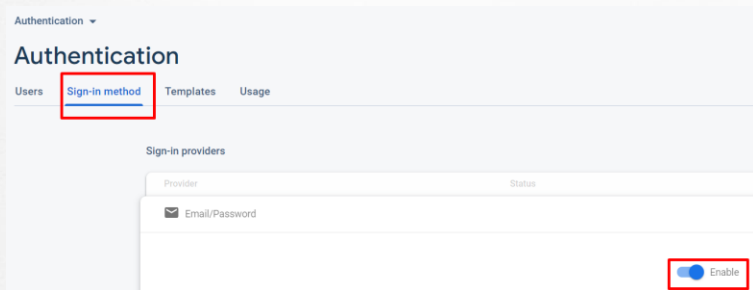
authentication-98e29-default-rtdb: null + ×

- ▶ In a Realtime Database we can tweak the rules to control how read and write access is rendered. If for example we have a data node members in our database we can set write and read rules for that part of the database. Auth is a special variable available in the Firebase rules which will not be null if there is a valid token.

```
{
  "rules": {
    "members": {
      ".read": true,
      ".write": "auth != null"
    },
    ".read": true,
    ".write": true
  }
}
```

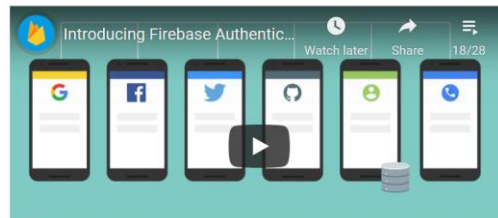
Firebase Authentication

- ▶ Firebase Console – Authentication – Get Started - Set Up a sign-in method
- ▶ Enable Email/Password
- ▶ With the above Firebase will support creating users with email and password.
- ▶ Firebase Authentication aims to make building secure authentication systems easy - [Link](#)



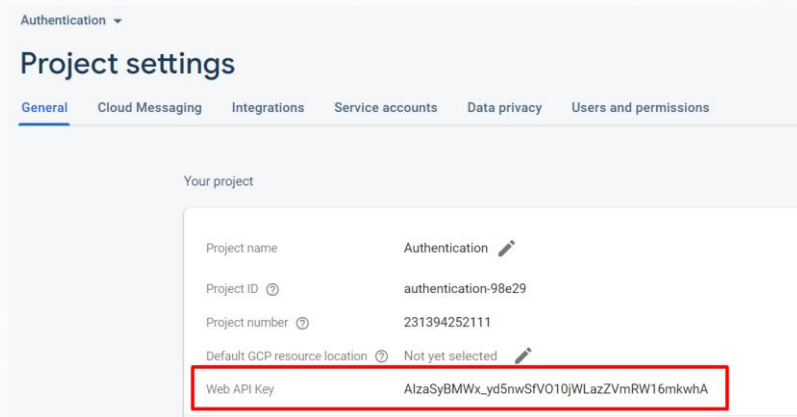
Easy sign-in with any platform

Firebase Authentication aims to make building secure authentication systems easy, while improving the sign-in and onboarding experience for end users. It provides an end-to-end identity solution, supporting email and password accounts, phone auth, and Google, Twitter, Facebook, and GitHub



Firebase Auth REST API

- ▶ The Firebase Auth backend through a REST API. This can be used for various operations such as creating new users, signing in existing ones and editing or deleting these users. The Web API_KEY can be obtained on the project settings page
- ▶ <https://firebase.google.com/docs/reference/rest/auth>



Exercise

- ▶ Research how you would implement authentication in a VUE application using a RealTime Database.
- ▶ In your assignment you are required to have users being able to register through a registration form and then a login form so that they can access different areas of the website which are not accessible to normal users.

References

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Any questions?

THANK YOU

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